

Process Characterization Report

A-Mab Drug Substance — Low-pH Viral Inactivation (Step 6) — PCR-006

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

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Field	Value
Document ID	PCR-006
Document class	Process Characterization Report
Title	Process Characterization Report: Low-pH Viral Inactivation (Step 6)
Product	A-Mab
Version	1.0
Effective date	2026-07-24
Status	Approved (synthetic)
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor / sites	Novacyte Biologics; Novacyte Biologics — Cambridge, MA (Development) → Novacyte Biologics — Grafton, WI (Commercial DS)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AMV, analytical method validation; ANOVA, analysis of variance; BLA, Biologics License Application; CCD, central composite design; Cpk, one-sided process capability index; CPP, critical process parameter; CQA, critical quality attribute; DCS, distributed control system; DoE, design of experiments; DS, drug substance; GPP, general process parameter; HMW, high molecular weight; icIEF, imaged capillary isoelectric focusing; LRF, log reduction factor; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; PAR, proven acceptable range; PRESS, predicted residual error sum of squares; RSM, response surface methodology; SDM, scale-down model; SEC-HPLC, size exclusion high performance liquid chromatography; SOP, standard operating procedure; TCID₅₀, fifty per cent tissue culture infectious dose; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

Executive summary

Low-pH viral inactivation is Step 6 of the A-Mab drug substance process and the first dedicated viral clearance step of the purification train. The Protein A eluate is titrated to a low pH, held for a defined time at controlled temperature, and neutralised before cation exchange chromatography (Step 7). Enveloped virus is inactivated by acid denaturation of the viral envelope. The step contributes 7.10 log₁₀ to the cumulative XMuLV clearance of 18.87 log₁₀ claimed for the process, which is the largest single contribution of the 3 credited steps (Table 8.2). The studies were executed on a qualified small-scale spiking model under the approved protocol PCP-006. They follow the lifecycle approach to process validation (U.S. Food and Drug Administration 2011), the enhanced development approach of ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023b, 2012), and the viral safety framework of ICH Q5A(R2) (International Council for Harmonisation 2023a).

The step has 4 parameters, 3 of which were characterized in a multivariate design (Table 4.1). Inactivation pH is classified as a critical process parameter. Hold time and temperature are well-controlled critical process parameters, and A-Mab concentration is a general process parameter. A two-level screening design of 11 runs identified pH, hold time and temperature as active on the log reduction factor, and a face-centred central composite design of 18 runs provided the response-surface models. The screening design identifies the active factors. The response-surface model is the predictive model and the basis of the operating region.

Across the characterized ranges the pool aggregate stays below 5 % and the acidic charge variants below 40 %, at every combination of the factors studied. The log reduction factor meets the required step contribution of 4.93 log₁₀ everywhere except at one corner of the characterized cube, where the highest pH is combined with the shortest hold and the lowest temperature. At that corner, 2 executed runs returned log reduction factors below the required contribution, the lowest of them 4.81 log₁₀. That corner is excluded from the operating region. The normal operating ranges lie well inside the region that remains, and at their least favourable corner the model predicts 6.43 log₁₀, a margin of 1.50 log₁₀ over the requirement.

At commercial scale the cumulative XMuLV clearance has a mean of 19.51 log₁₀ against a requirement of at least 16.7 log₁₀, a margin of 2.81 log₁₀, and a one-sided Cpk of 1.53 (Table 8.1). It is the tightest of the attributes this step governs, and the capability rests on the other credited steps as well as on this one. The study recorded 2 deviations during execution. Both were investigated and dispositioned as retained, and neither altered a parameter classification or the reported operating region.

No clearance of non-enveloped virus is claimed for this step, and its MVM contribution is reported as 0.0 log₁₀ (Table 8.2). The cumulative XMuLV claim also rests on the

anion exchange step (PCR-008) and on the small-virus retentive filtration step (PCR-009), which act by different mechanisms. The step assures a defined enveloped-virus inactivation increment and a bounded product quality cost. It does not by itself assure the viral safety of the drug substance. This report rolls up into PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody whose primary mechanism of action is antibody-dependent cellular cytotoxicity. It is produced by fed-batch cell culture and purified by a platform train of Protein A capture, low-pH viral inactivation, cation exchange chromatography, anion exchange chromatography, small-virus retentive filtration and ultrafiltration with diafiltration (Steps 5–10). The train is the one described in the A-Mab case study (CMC Biotech Working Group 2009), and this report covers the second of its downstream steps.

The step is a batch hold. The Protein A eluate is titrated with acid to the target pH in a mixed vessel, held for the target time under temperature control, and then neutralised to the pH required by the cation exchange load. Inactivation of enveloped virus proceeds by acid-driven denaturation of the envelope glycoproteins, which is a kinetic process. Its extent therefore depends on the pH reached, on how long the material is held there, and on the temperature at which it is held. The same conditions act on the antibody, where partial acid unfolding produces high molecular weight species (HMW) and asparagine deamidation over the hold raises the acidic charge variant level. The step is a balance between these two effects, and characterizing it means bounding both.

The step sets the XMuLV clearance credited to it in the modular viral safety claim (Table 2.1). It does not form the aggregate or the charge variant attributes, which are established in the production bioreactor (PCR-003). It adds to both, and aggregate is subsequently reduced at cation exchange (PCR-007). Clearance of host cell protein by precipitation at low pH is not claimed here, since it is not reliably predictable across feed states.

1.2 Objectives

The objectives of the characterization were as follows.

- (i) Quantify the relationship between the operating parameters of the step and the XMuLV log reduction factor, the pool aggregate level and the acidic charge variant level.
- (ii) Define the multivariate operating region within which all three responses meet their acceptance criteria.

- (iii) Establish the proven acceptable range of each parameter for each attribute it governs, both with the other parameters held fixed and with them varying within their normal operating ranges.
- (iv) Classify each parameter against the decision logic of SOP-4001.
- (v) Justify the ranges to be run in Stage 2 and the contribution this step makes to the control strategy for the drug substance.

1.3 Regulatory and scientific basis

The study is a Stage 1 process design activity in the sense of the lifecycle approach to process validation (U.S. Food and Drug Administration 2011). Its record is the evidence that the commercial ranges were established from designed experiments on a qualified model, and not from a small number of engineering runs. The enhanced development approach of ICH Q8 and the risk framework of ICH Q9 apply throughout, and the parameter classification treats criticality as a continuum (International Council for Harmonisation 2023b). ICH Q11 frames the link from the studies to the control strategy (International Council for Harmonisation 2012).

Viral safety is evaluated under ICH Q5A(R2) (International Council for Harmonisation 2023a). That guideline sets the framework used here in three ways. Clearance is claimed on a modular basis, so that each step is studied independently and the increments are summed. Clearance data are generated at small scale with a deliberately spiked feed, because virus cannot be introduced into a manufacturing process. Steps are credited only where the mechanism of removal or inactivation is understood and the operating conditions that deliver it are controlled. The A-Mab case study provides the process model, the attribute framework and the risk methodology (CMC Biotech Working Group 2009).

This report is one document of a cross-referenced package. Its risk basis, its paired protocol and the report it rolls up into are listed in Table 1.1.

Table 1.1: Related documents in the A-Mab characterization package.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-006	Process Characterization Plan (Low-pH Viral Inactivation (Step 6))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The low-pH hold is a platform step with a long performance history. It has been used to manufacture three previously licensed humanized IgG1 antibodies (X-Mab, Y-Mab and Z-Mab), and its operating conditions have remained essentially unchanged across those products and throughout A-Mab development. The mechanism of inactivation is not product specific, since it depends on the pH and temperature of the solution and on the exposure time, so the knowledge gained on the platform transfers directly to A-Mab. Characterization therefore confirms and bounds the step for this molecule. It is not a search for the operating conditions.

Prior knowledge sets a clear mechanistic expectation, and the study was designed to test it. Log reduction should rise as pH falls, as the hold lengthens and as the temperature rises, because all three increase the extent of acid denaturation over the hold. Aggregate should rise with hold time and with temperature for the same reason acting on the antibody, and should be largely independent of pH within the narrow band studied. Acidic charge variants should rise with hold time, since deamidation accumulates with exposure. If the data contradict any of these expectations, the platform justification for the step would have to be re-examined.

The same prior knowledge sets the edges of the characterized ranges. The low edge of the pH range sits above the region in which antibody precipitation has been observed on the platform, so that precipitation could not confound the study. The high edge sits above the pH at which inactivation of enveloped virus has been shown to be reliable within a short hold, which is what makes the upper part of the pH range the interesting part. The hold time range extends to 3 times its lower edge, and the temperature range spans the controlled room-temperature band available in the manufacturing suite.

2.2 Quality attributes in scope

The step governs 3 drug substance attributes, and they differ widely in criticality (Table 2.1). Criticality was assigned by the Tool #1 method of the case study (impact multiplied by uncertainty), and attributes scoring high or very high are called critical (CMC Biotech Working Group 2009).

Table 2.1: Quality attributes governed by the low-pH viral inactivation step, with acceptance criteria and criticality.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — XMuLV (enveloped)	viral safety	16.7-30 log10 (cumulative)	VH	20
Aggregates (HMW)	size variant	0-5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18-40 % (CEX/icIEF)	VL	4

XMuLV clearance is the attribute this step sets. It is of very high criticality, and its acceptance criterion is cumulative across the credited steps, which is why the criterion in Table 2.1 is larger than any single step can deliver. Aggregate is of high criticality and carries the highest Tool #1 score of the three. It is formed upstream, added to here, and polished downstream, so this step's obligation is to bound its own increment. Acidic charge variants are of very low criticality, and the attribute is carried in scope because the hold adds to the level formed upstream and that increment has to be bounded.

2.3 Risk-based prioritization and parameter selection

The scope of the study was set by RA-001, which ranked every parameter of the step by its potential impact on a CQA and by its potential to interact with other parameters. The ranking sent 3 of them to multivariate study. Inactivation pH carries the largest expected effect on log reduction and the narrowest operating band. Hold time drives log reduction, aggregate and deamidation at once, which makes it the parameter most likely to force a compromise. Temperature acts on the same kinetics as hold time and was expected to interact with it.

A-Mab concentration was assigned to univariate assessment. Its plausible mechanism is a matrix effect on the acid titration and on the local ionic environment, which prior knowledge places well below the three kinetic factors in magnitude, and no interaction with them was expected. Studying it one factor at a time over its full range was judged sufficient. The ranges taken forward for the multivariate factors are wider than their normal operating ranges (Table 4.1), which is deliberate. A range that only covers the control band cannot distinguish a robust parameter from an unstudied one, and it leaves no room for movement within the design space later in the lifecycle.

3 Materials and methods

The step was executed on a qualified scale-down model under the controlled documents listed in Table 3.1. The procedures cover model qualification, the design and analysis of the DoE studies, the operation of the step itself, and the parameter classification logic (SOP-4001) applied in the classification section below.

Table 3.1: Controlled documents and validated analytical methods applied to this step.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2009	Operation of the Low-pH Viral Inactivation Step	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

3.1 Scale-down model and its qualification

The scale-down model is a jacketed, mixed vessel operated with the same pH, temperature and hold profile as the commercial step. The variables that govern the inactivation are scale-independent, because the hold is a homogeneous liquid-phase reaction with no transport limitation. The model therefore reproduces the commercial step by matching the acid titration path, the target pH, the hold time, the temperature and the neutralisation, and by using representative Protein A eluate as the feed. Mixing was set to give the same time to pH homogeneity as the commercial vessel, so that no part of the batch experiences a shorter effective hold than the record states.

Qualification followed SOP-1001, under which the model was operated at the target conditions and its input and output attributes were compared with the commercial-scale batch data (pool aggregate, acidic charge variants, protein concentration and step recovery). No difference of practical consequence was found, and the model is accepted as representative for the definition of operating ranges. Clearance claims rest on the same qualification with one addition required by ICH Q5A(R2). Virus can

only be introduced at small scale, so the spiking model must also be shown to hold the feed matrix, the spike volume and the neutralisation step within the limits under which the clearance is claimed (International Council for Harmonisation 2023a). The spike volume was kept small enough that the acid titration path of the run was set by the eluate and not by the spike.

Replicate centre-point runs were included in both designs, 3 in the screening design and 4 in the response-surface design. They provide the pure-error estimate against which lack of fit is tested, and they also give a direct measure of the reproducibility of the model and the assays.

3.2 Operation

Each run started from a representative Protein A eluate held at the platform target concentration. The eluate was titrated with acid to the run's target pH under mixing, and the pH was confirmed by an independent probe (a second, separately calibrated instrument) before the hold clock was started. The material was then held at the run's target temperature for the run's target time, sampled, and neutralised. Samples for infectivity titration were taken from the spiked hold vessel immediately before neutralisation, and product quality samples were taken from the neutralised pool.

Inputs other than the three factors were controlled to their platform specifications and were not varied. Buffer identity, acid and base titrant concentration, vessel geometry and mixing rate all fall into this group. These are identity and quality controlled inputs, and they are verified at use. They were not characterized in this study.

3.3 Analytical methods

XMuvLV infectivity was determined by TCID₅₀ titration under AMV-3017. The method has a validated intermediate precision of 0.30 log₁₀, a quantitation limit of 1.5 log₁₀ TCID₅₀/mL and an accuracy of 96 %. Method validation attributes 28.0% of the total observed variance in a reported titre to the assay. The log reduction factor is the difference between the titre of the spiked load and the titre of the post-hold sample, so a reported log reduction factor carries the uncertainty of two titrations and not of one.

Pool aggregate was determined by SEC-HPLC under AMV-3011 and reported as a percentage of high molecular weight species. Acidic charge variants were determined by icIEF under AMV-3013 and reported as a percentage of the total charge variant profile. The techniques, analytes and reportable values are summarized in Appendix D. Full method performance is held in the validation reports referenced in Table 3.1, and is not reproduced here.

3.4 Sampling plan

Every run was sampled twice for infectivity, once from the spiked load before acidification and once from the hold vessel at the end of the hold. Product quality samples were drawn from the neutralised pool of every run and assayed for aggregate and charge variants. The centre-point runs were sampled in the same way as the factorial and axial runs, so that the replicate estimate of variance covers sampling and assay variation as well as run-to-run process variation.

3.5 Statistical methods

Factors were coded so that the centre of each characterization range is zero and the two edges are minus one and plus one. The screening design was analysed by ordinary least squares (OLS) with all main effects and all two-factor interactions, and an effect is reported as twice the coded coefficient, which is the change in the response across the full range of that factor. The response-surface design was analysed with the full quadratic model in the same factors. The significance level is 0.05 throughout.

Model adequacy was judged on four grounds. The coefficient of determination and its adjusted form measure how much of the observed variation the model explains. The predicted form, computed from the PRESS statistic, measures how well the model predicts a run that was left out of the fit, and it is the most demanding of the three measures. The overall F test determines whether the model explains more than noise. Lack of fit is tested by comparing the residual variation that the model cannot explain with the pure error estimated from the replicate centre points, and a significant lack of fit would mean that the fitted form is too simple for the data. A factor with no significant term in either analysis is treated as having no demonstrated effect over the range studied.

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches with every parameter of the train drawn inside its normal operating range, and is reported as a one-sided Cpk against the acceptance criterion of each attribute.

4 Study design

The design resolves the kinetic factors in two stages. The screening stage determines which factors and interactions are active, and the response-surface stage adds the curvature terms needed to describe the region and to predict inside it. The factors, their set-points, their normal operating ranges and the ranges characterized are given in Table 4.1, and the coded letters used throughout the results are given in Table 4.2.

4.1 Factors, ranges and the knowledge space

Table 4.1: Parameters of the low-pH viral inactivation step, with set-points, normal operating ranges, characterization ranges and final classification.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Inactivation pH	pH	3.5	3.4–3.6	3.2–4	CPP	multivariate
Hold time	min	90	60–120	60–180	WC-CPP	multivariate
Temperature	°C	21	19–23	15–25	WC-CPP	multivariate
A-Mab concentration	g/L	20	5–35	5–35	GPP	univariate

Table 4.2: Coded factor legend used in the effect and coefficient tables.

Code	Factor	Unit	Studied in
A	Inactivation pH	pH	screening + RSM
B	Hold time	min	screening + RSM
C	Temperature	°C	screening + RSM

The characterization ranges in Table 4.1 are the knowledge space of the step, and each was justified before execution. The pH range is the widest of them relative to its normal operating range, because pH is the parameter with the largest expected effect and the narrowest control band, and because the upper part of the range is where the step is expected to lose effectiveness. The hold time range starts at the shortest hold that would be accepted and extends well past the control band, so that a curtailed hold and a prolonged hold caused by a downstream delay are both covered.

The temperature range spans the full controlled room-temperature band. A-Mab concentration was studied across the whole range over which the Protein A eluate is delivered (PCR-005), which is why its characterization range and its normal operating range coincide.

4.2 Screening design

The screening study was a two-level full factorial in the three multivariate factors with 3 centre points, giving 11 runs in total. A full factorial was chosen over a fraction because the factors are few enough that every two-factor interaction (AB, AC and BC) can be estimated independently of the main effects. The design resolves every main effect and every two-factor interaction, and the centre points allow curvature to be detected as an aggregate departure from the plane. The runs were executed in randomised order, and the full matrix and the measured responses are given in Appendix A.

4.3 Response-surface design

The response-surface study was a face-centred central composite design in the same factors, with 6 axial runs added to the 8 factorial corners and 4 centre points, giving 18 runs. The axial points sit on the faces of the cube (the face-centred convention), so the design stays inside the characterized ranges and does not require any factor to be set beyond an edge that has been justified. Every screening factor was carried into the response-surface stage, because each was active on at least one response. The design adds the quadratic terms to the screening model and supports prediction inside the region. The full matrix and the measured responses are given in Appendix B.

4.4 Univariate assessment

A-Mab concentration was assessed one factor at a time across its full range with the kinetic factors at their set-points. It was excluded from the multivariate designs for the reason given in the risk section above. Its expected mechanism is a matrix effect on the acid titration, and it does not enter the inactivation kinetics as a term of its own. Adding it to the multivariate design would have doubled the run count for a factor with no expected interaction. The outcome of the assessment is reported with the parameter classification below.

5 Results

5.1 Centre-point performance and reproducibility

The step is highly reproducible in the two product quality responses and considerably less so in the log reduction factor. The 4 replicate centre points of the response-surface design give the reproducibility shown in Table 5.1.

Table 5.1: Centre-point reproducibility of the response-surface design, by response.

Response	n	Mean	SD	%CV
Aggregates (HMW, %)	4	2.01	0.0169	0.845
XMuLV LRF ($\log_{10}$)	4	6.71	0.381	5.67
Acidic variants	4	20.7	0	0

The log reduction factor is by far the least precise response, at 5.7 % of its mean, against 0.84 % for the pool aggregate. In absolute terms the replicate standard deviation of the log reduction factor is 0.38 $\log_{10}$, and the validated intermediate precision of the infectivity assay is 0.30 $\log_{10}$ for a single titration. A reported log reduction factor is the difference of two titrations, so the scatter seen between replicate holds is consistent with analytical variation alone. The screening centre points behave the same way, at 3.5 % across 3 runs.

The acidic charge variant response shows no replicate variation at all. Its centre-point standard deviation is zero to the resolution of the reported result, and the same value was returned by every centre-point run in both designs. This has a consequence for the statistics that is dealt with in the response-surface subsection below, and it should be read as a property of how the attribute responds to this step, not as evidence of unusual analytical precision.

5.2 Screening: factor effects

The screening study identified pH, hold time and temperature as active, and found no significant interaction between any pair of them. Model adequacy for every response is given in Table 5.2, and the effect estimates for the two informative responses in Table 5.3 and Table 5.4. The full effect tables for all three responses are in Appendix C.

Table 5.2: Screening model adequacy by response.

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Aggregates (HMW, %)	11	0.992	0.98	0.828	82	0.000386	0.0333
XMuLV LRF (log ₁₀)	11	0.982	0.956	0.819	37.2	0.00182	0.265
Acidic variants	11	1	1	1	3.93e+26	1.73e-53	1.16e-14

Both informative models explain the screening data well. The pool aggregate model reaches an adjusted coefficient of determination of 0.980 and the log reduction model 0.956, and both are significant overall ($p = 0.0018$ for the log reduction factor). The acidic charge variant row of Table 5.2 should not be read as a model result. The response is an exact linear function of hold time in these data, so the residual variation is at the level of floating-point arithmetic and the F statistic and p-value that follow from it carry no information.

Table 5.3: Largest screening effects on the XMuLV log reduction factor. Effect is the change in the response across the full characterization range of the factor.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	-2.08	-1.04	0.0936	-11.1	0.000376	***
B	1.72	0.862	0.0936	9.2	0.000774	***
C	0.636	0.318	0.0936	3.4	0.0274	*
BC	0.238	0.119	0.0936	1.27	0.272	

pH is the dominant factor on the log reduction factor. Moving pH across its full characterization range changes the log reduction factor by -2.08 log₁₀ ($p = 0.0004$), and the sign is negative, so clearance falls as pH rises. Hold time is next at +1.72 log₁₀ across its range, and temperature is third at +0.64 log₁₀ ($p = 0.027$). All three effects have the direction prior knowledge predicted. The largest interaction, between hold time and temperature, is not significant ($p = 0.27$), which means the three factors act on the log reduction factor additively over the ranges screened.

Table 5.4: Largest screening effects on the pool aggregate level.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	0.471	0.236	0.0118	20	3.69e-05	***
C	0.221	0.11	0.0118	9.37	0.000723	***
AC	0.035	0.0175	0.0118	1.48	0.212	
AB	0.0285	0.0142	0.0118	1.21	0.294	

The pool aggregate is driven by hold time and temperature, and not by pH. Hold time changes the pool aggregate by +0.47 percentage points across its range and

temperature by +0.22 points ($p = 3.7\text{e-}05$ and $p = 0.0007$ respectively). The pH effect is +0.022 points and is not significant ($p = 0.40$). That null result matters to the design of the step, because it means the pH needed for inactivation can be chosen on viral clearance grounds alone, without trading it against aggregate, and pH remains in the knowledge space as a factor with no demonstrated effect on aggregate across the range studied.

Screening establishes which factors matter and gives a first estimate of how much. The magnitudes above are estimated from a design with two levels per factor, so they describe a plane through the region and cannot represent curvature. They are refined by the response-surface analysis that follows, which is the model used for prediction.

5.3 Response-surface models

The response-surface study confirms the screening result, adds curvature in pH on the log reduction factor, and gives two adequate predictive models. Model adequacy is summarized in Table 5.5.

Table 5.5: Response-surface model adequacy by response.

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Aggregates (HMW, %)	18	0.997	0.993	0.961	261	6.57e-09	0.0169
XMuLV LRF (log ₁₀)	18	0.896	0.779	0.579	7.67	0.00433	0.367
Acidic variants	18	1	1	1	5.99e+26	2.43e-106	8.61e-15

The pool aggregate model is strong on every measure, with a coefficient of determination of 0.997, an adjusted value of 0.993 and a predicted value of 0.961. The log reduction model is weaker and needs to be described carefully. It is significant overall ($p = 0.0043$) and its coefficient of determination is 0.896, but its predicted value falls to 0.579. The gap between those two figures is the signature of a response whose replicate scatter is large relative to its systematic variation, which is what the centre-point analysis above shows for this assay. The model is used to predict mean log reduction inside the characterized region. It is not used as an assurance statement about a single batch, and its residual standard error of 0.37 log₁₀ is the right scale on which to judge any predicted margin.

The acidic charge variant model is reported for completeness and carries no statistical weight. Its coefficient of determination is exactly one because the response is a deterministic linear function of hold time in these data. Its pure error is zero, so lack of fit cannot be tested, and its F statistic is a numerical artefact. The physical content of that row is simply that the acidic charge variant level rises with hold time and with nothing else in this step.

Table 5.6: Full quadratic coefficients for the XMuLV log reduction factor, on coded factors.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	6.91	0.144	47.9	3.99e-11	***
A	-0.612	0.116	-5.28	0.000747	***
B	0.605	0.116	5.22	0.000806	***
C	0.286	0.116	2.46	0.0391	*
AB	0.162	0.13	1.25	0.246	
AC	0.0691	0.13	0.533	0.608	
BC	-0.105	0.13	-0.812	0.44	
A ²	-0.513	0.223	-2.31	0.0501	
B ²	0.202	0.223	0.907	0.391	
C ²	0.152	0.223	0.684	0.513	

The coefficients in Table 5.6 reproduce the screening picture and add one term. The main effects are all significant, with pH at -0.61 log₁₀ per coded unit, hold time at +0.60 and temperature at +0.29. No two-factor interaction is significant, which confirms the additive behaviour seen in screening. The quadratic term in pH is -0.51 at $p = 0.050$, which is marginal. Taken at face value it means the gain in clearance flattens at the low-pH end of the range, so that lowering the pH below the set-point buys progressively less inactivation. That is consistent with a denaturation process approaching completion, and it is treated here as a suggestion supported by one borderline term. It is not presented as an established feature of the surface.

Table 5.7: Analysis of variance with the lack-of-fit partition for the XMuLV log reduction factor.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	9.27	9	1.03	7.67	0.00433
Residual	1.07	8	0.134	nan	nan
Lack of fit	0.64	5	0.128	0.884	0.58
Pure error	0.435	3	0.145	nan	nan

Lack of fit is not significant for the log reduction factor ($F = 0.88$, $p = 0.58$), so the quadratic form is adequate for the data and the unexplained variation is consistent with pure error. Table 5.7 also shows why the model is hard to sharpen. The lack-of-fit mean square is 0.128 and the pure error mean square is 0.145, so the variation the model leaves unexplained is no larger than the scatter between replicate holds. Adding terms would not improve prediction. The corresponding coefficient and variance tables for the pool aggregate are given in Table 5.8 and Table 5.9.

Table 5.8: Full quadratic coefficients for the pool aggregate level, on coded factors.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	2.01	0.00667	301	1.64e-17	***
A	-0.00244	0.00536	-0.455	0.661	
B	0.236	0.00536	44.1	7.76e-11	***
C	0.107	0.00536	20	4.01e-08	***
AB	-0.00133	0.00599	-0.221	0.83	
AC	0.0133	0.00599	2.21	0.0578	
BC	0.00837	0.00599	1.4	0.2	
A ²	-0.00247	0.0103	-0.24	0.816	
B ²	0.000627	0.0103	0.0609	0.953	
C ²	0.0125	0.0103	1.21	0.261	

Table 5.9: Analysis of variance with the lack-of-fit partition for the pool aggregate level.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	0.676	9	0.0751	261	6.57e-09
Residual	0.0023	8	0.000287	nan	nan
Lack of fit	0.00144	5	0.000287	1	0.535
Pure error	0.000861	3	0.000287	nan	nan

The pool aggregate surface is close to a plane in hold time and temperature. Hold time contributes +0.236 percentage points per coded unit and temperature +0.107, both far outside the noise. The interaction between pH and temperature is the only other term that comes close to significance ($p = 0.058$), and its coefficient is +0.013 points, which is too small to matter over any part of the region. Lack of fit is not significant ($p = 0.53$).

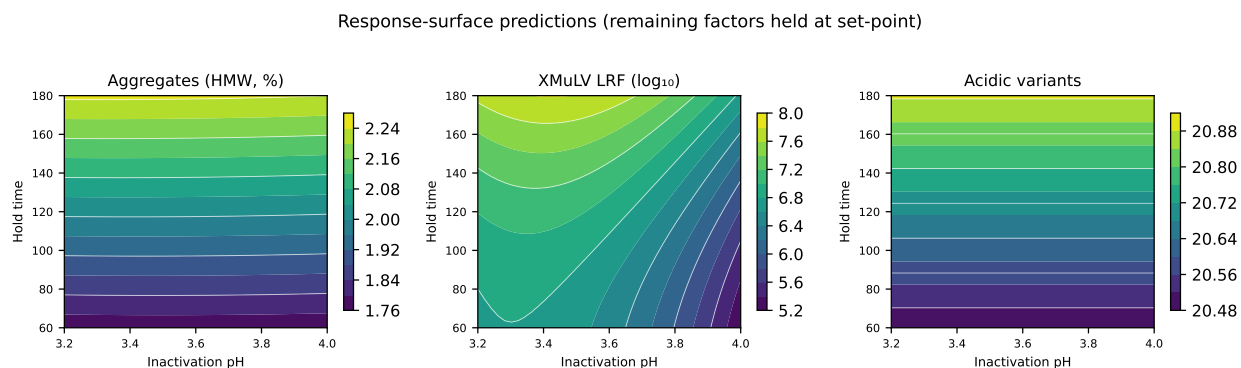


Figure 5.1: Response-surface predictions over inactivation pH and hold time for the three responses, with temperature held constant.

The surfaces in Figure 5.1 show the shape of the region. The log reduction contours run diagonally, because a rise in pH can be offset by a longer hold, and the corner at the highest pH and the shortest hold is the only part of this plane where clearance approaches its limit. The aggregate contours run horizontally, because they depend on hold time and not on pH. The two surfaces therefore constrain the region in different directions, and a long hold that helps clearance costs aggregate.

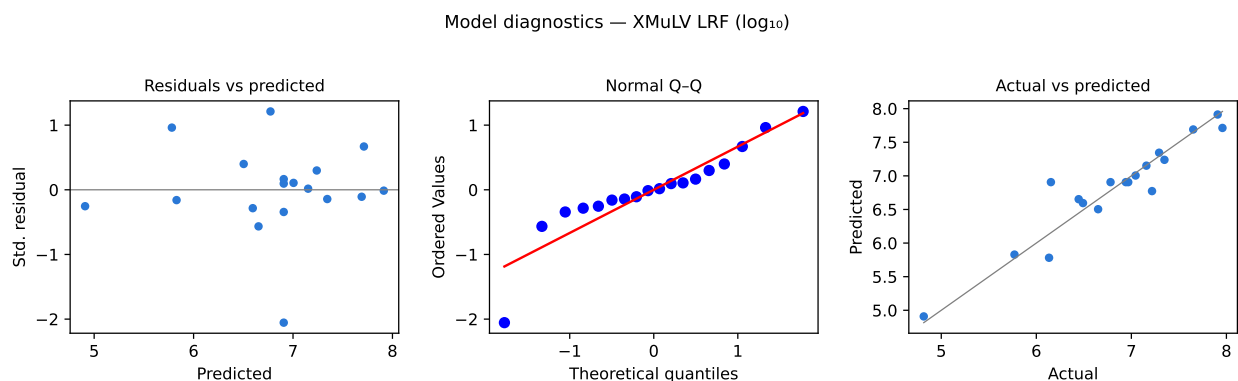


Figure 5.2: Model diagnostics for the XMuLV log reduction factor: standardized residuals against predicted values, normal quantile plot, and actual against predicted values.

The diagnostics for the log reduction model are shown in Figure 5.2. The residuals show no structure against the predicted value and no trend in spread, the quantile plot is close to a straight line, and the actual values scatter about the identity line without systematic curvature. The scatter about that line is wide, which is the same finding as the low predicted coefficient of determination and has the same cause. The corresponding diagnostics for the pool aggregate model are given in Appendix C.

5.4 Mechanistic interpretation

The surfaces have the shape the underlying chemistry predicts, and that agreement is part of the evidence for the step. Acid inactivation of an enveloped virus is a denaturation reaction. Its rate rises as the pH falls further below the stability limit of the envelope glycoproteins and as the temperature rises, and the extent achieved is the rate integrated over the hold. A model in which pH, hold time and temperature act additively on a logarithmic measure of survival is what that mechanism predicts, and it is what the coefficients show. No coupling between the three factors was demonstrated over the ranges studied, so the main effects and the quadratic term in pH describe the surfaces adequately. Any curvature in the boundary of the operating region comes from that quadratic term in pH, and not from two factors acting together.

Aggregate formation is driven by the same acid exposure acting on the antibody instead of on the virus, and the species formed are dimer and higher order (SEC-HPLC). It responds differently, though, tracking hold time and temperature while staying flat in

pH across the band studied. The likely reason is that the antibody is already below the pH of its acid-labile conformational transition even at the top of the range studied, so a further reduction in pH changes the exposure very little. That is consistent with the platform history, in which the low-pH hold has been operated across this band for three prior products without an aggregate problem.

Deamidation behaves as a simple time-dependent chemical modification. The acidic charge variant level rises with hold time alone, and across the whole characterization range for hold time it rises by 0.40 percentage points, from 20.49 % at the shortest hold to 20.89 % at the longest, against an acceptance limit of 40 %. The attribute is real, the mechanism is understood, and the magnitude is negligible against its criterion.

The practical consequence of these three mechanisms is that the step cannot be described by independent ranges on its parameters. Clearance and aggregate move in opposite directions with hold time, and clearance depends on the combination of pH, hold time and temperature rather than on any one of them. A multivariate region is therefore needed, and it is defined in the next section.

6 Design space

The design space of the step is the part of the characterized cube in which the predicted XMuLV log reduction factor meets the required step contribution and both product quality responses stay within their drug substance limits. The two product quality constraints are inactive. Over the whole characterized cube the fitted model predicts a pool aggregate of at most 2.38 % against a limit of 5 %, and an acidic charge variant level of at most 20.89 % against a limit of 40 %. The highest pool aggregate measured in either design was 2.44 % (Table 14.2 and Table 14.4), which is also far below the limit. Neither attribute constrains the region anywhere inside the ranges studied. The binding constraint is viral clearance.

The clearance constraint binds at one corner. The least favourable combination for clearance is the highest pH studied with the shortest hold and the lowest temperature, and there the fitted model predicts $4.91 \log_{10}$ against a required step contribution of $4.93 \log_{10}$. The corner fails in the model, and it failed in the executed data. At that corner, 2 runs returned measured log reduction factors below the requirement, the lowest being $4.81 \log_{10}$. The predicted shortfall of $0.02 \log_{10}$ is small in absolute terms, and it is much smaller than the residual standard error of the model, which is $0.37 \log_{10}$. A shortfall inside the noise cannot be argued away as noise, because the same argument would prevent the corner from being declared acceptable. The corner is excluded.

Holding the shortest hold time and the lowest temperature, the model meets the requirement up to pH 3.99, so the excluded region is a thin slice at the top of the pH range and only when both of the other factors are also at their least favourable edge. At the set-point hold time and temperature the requirement is met across the whole pH range. The design space is therefore the characterized cube less that corner, and its principal plane is pH against hold time, which is the plane shown in Figure 5.1. Temperature shifts the surface without changing its shape.

The normal operating ranges lie well inside this region. At the set-point the model predicts $6.87 \log_{10}$, and at the least favourable corner of the normal operating ranges it predicts $6.43 \log_{10}$, a margin of $1.50 \log_{10}$ over the requirement. The normal operating ranges sit inside the design space, and the design space sits inside the knowledge space bounded by the characterization ranges of Table 4.1.

Three bounds apply to this claim. It holds only over the characterized ranges of the three factors and only for feed material within the platform specification for Protein A eluate, since the feed was not varied in these designs. It is a claim about mean predicted responses from a fitted model, so the reliability at the boundary of the region is lower than it is at the centre, and it should not be read as an assurance figure for an individual batch. It rests on the qualified scale-down model, and, as for every viral clearance claim, the clearance itself has been demonstrated only at small

scale. Movement within the region is not a change in the regulatory sense of ICH Q8, although it remains subject to the pharmaceutical quality system. Movement outside the region requires a change control assessment and, where the excluded corner is concerned, new data.

7 Proven acceptable ranges

The design space is a multivariate statement. The proven acceptable ranges complement it with a per attribute, per parameter view, and they answer a narrower question about how far one parameter can be moved before the attribute it governs leaves acceptance. The acceptance basis differs between the attributes. For the pool aggregate and the acidic charge variants it is the drug substance specification given in Table 2.1. For XMuLV it is the contribution this step must make to the cumulative claim, which is the cumulative requirement less the clearance credited to the other steps, and it is $4.93 \log_{10}$.

Two analyses were run for every attribute and parameter pair, and both are given in Table 7.1. The first holds the other parameters fixed and scans the parameter of interest across its characterization range on a grid, using the fitted response-surface model. The second varies the other parameters within their normal operating ranges by Monte-Carlo of the same model, and takes the range over which the 95 % predictive interval stays within acceptance. The second is a robustness criterion and is the narrower of the two by construction.

Table 7.1: Proven acceptable ranges by attribute and parameter, with the characterization range for comparison.

CQA	Parameter	Char. range	Unit	PAR (set-point)	PAR (NOR)
Aggregates (HMW, %)	Inactivation pH	3.2-4	pH	3.2-4	3.2-4
Aggregates (HMW, %)	Hold time	60-180	min	60-180	60-180
Aggregates (HMW, %)	Temperature	15-25	°C	15-25	15-25
XMuLV LRF ($\log_{10}$)	Inactivation pH	3.2-4	pH	3.2-4	3.2-3.96
XMuLV LRF ($\log_{10}$)	Hold time	60-180	min	60-180	60-180
XMuLV LRF ($\log_{10}$)	Temperature	15-25	°C	15-25	15-25
Acidic variants	Inactivation pH	3.2-4	pH	3.2-4	3.2-4
Acidic variants	Hold time	60-180	min	60-180	60-180
Acidic variants	Temperature	15-25	°C	15-25	15-25

For the pool aggregate and the acidic charge variants the two analyses agree with each other and with the characterization range, for every parameter. The whole studied range of every parameter is proven acceptable for those two attributes, and it stays acceptable when the other two parameters move within their normal operating ranges. That is a strong robustness result, and it is bounded by the ranges in the third column of Table 7.1. Nothing is claimed beyond them.

For XMuLV clearance the two analyses agree for hold time and for temperature, and they differ for pH. The whole hold time range and the whole temperature range are proven acceptable under both analyses, which is expected, since both factors raise clearance as they increase and their lower edges were included in the region only in combination with a high pH. The pH range narrows under the second analysis, from the full characterization range to an upper edge of 3.96. XMuLV clearance is the binding attribute, and the residual margin is 0.36 pH units above the upper edge of the normal operating range for pH. In other words, pH would have to be run above its control band, with hold time and temperature simultaneously unfavourable inside their own control bands, before the predictive interval touched the requirement.

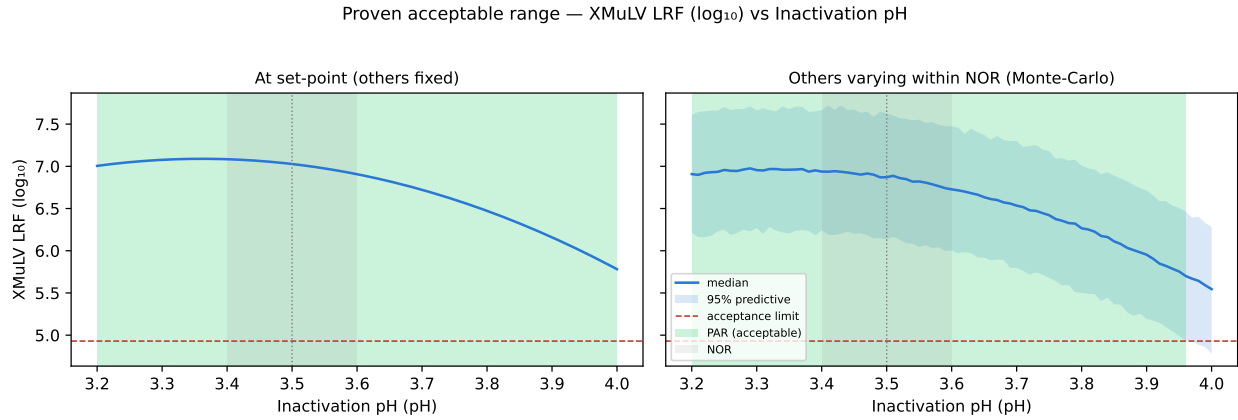


Figure 7.1: Proven acceptable range of inactivation pH for the XMuLV log reduction factor. The acceptable region is shaded green, the acceptance limit is drawn, and the set-point and normal operating range are marked.

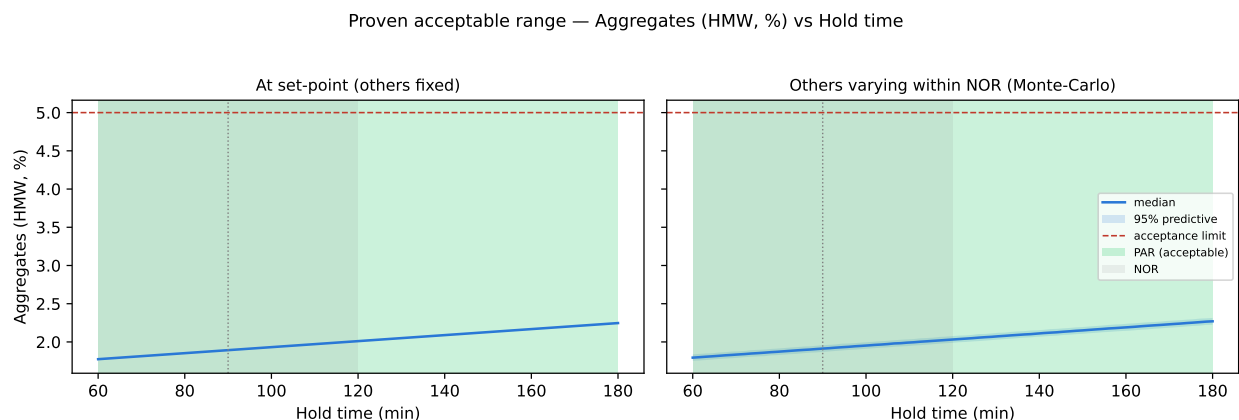


Figure 7.2: Proven acceptable range of hold time for the pool aggregate level.

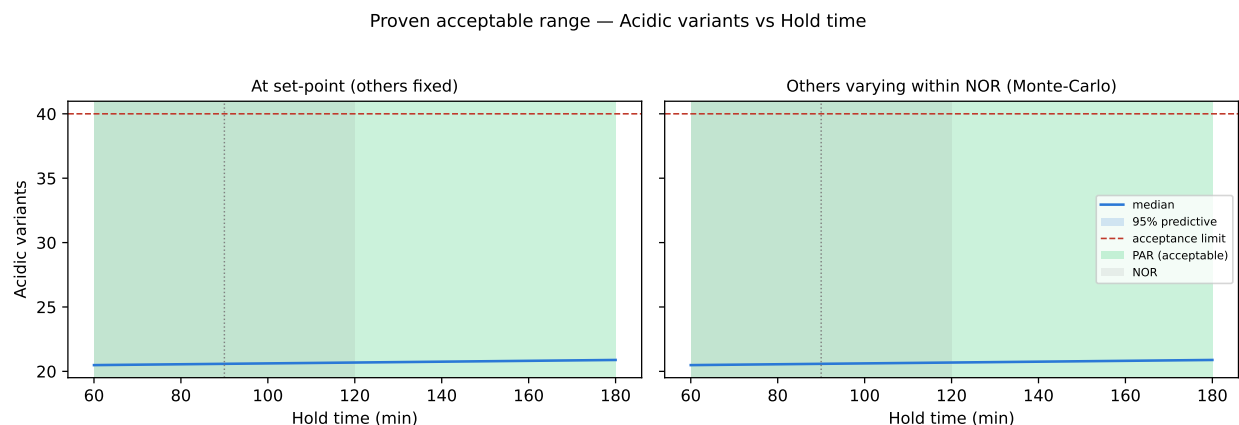


Figure 7.3: Proven acceptable range of hold time for the acidic charge variant level.

Each governed attribute is plotted against its governing parameter, which is the parameter with the largest main effect on it in the response-surface model. Figure 7.1 shows XMuLV clearance against pH, and the two panels show how much the acceptable region contracts when hold time and temperature are allowed to vary within their control bands. Figure 7.2 and Figure 7.3 show the pool aggregate and the acidic charge variants against hold time, and in both cases the acceptable region covers the whole studied range with a wide margin to the limit.

These ranges and the design space answer different questions, and the two are not interchangeable. A proven acceptable range is a one-dimensional statement about a single parameter. The design space is the multivariate region, and it is the smaller claim wherever the parameters interact or, as here, wherever the attribute depends on a combination of them. The pH range is the only case in this step where a univariate range binds at all, and even there the corner excluded from the design space is excluded by the combination of all three factors and not by pH alone. The binding constraint on the operating region is therefore the design-space corner described above, not any univariate proven acceptable range. Both statements are bounded by the characterization ranges and by the fitted model.

8 Process capability and robustness

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches with every parameter of the train drawn inside its normal operating range (not only the parameters of this step). The results for the attributes this step governs are given in Table 8.1.

Table 8.1: Commercial-scale capability for the attributes governed by this step.

CQA	Crit.	Spec	Acceptance	Mean $\pm$ SD	Cpk
Aggregates (HMW)	H	upper	0-5	0.753 $\pm$ 0.088	16.1
Acidic charge variants (deamidation)	VL	upper	18-40	22.3 $\pm$ 1.4	4.26
Viral clearance — XMuLV (enveloped)	VH	lower	16.7-30	19.5 $\pm$ 0.61	1.53

XMuLV clearance is the tightest of them by a wide margin, at a one-sided Cpk of 1.53. Its simulated mean is 19.51 $\log_{10}$ against a requirement of at least 16.7 $\log_{10}$, so the margin to the limit is 2.81 $\log_{10}$. The pool aggregate and the acidic charge variants are far from their limits, at Cpk values of 16.1 and 4.26 respectively, and the drug substance aggregate mean of 0.75 % sits well below its ceiling of 5 %.

None of these figures belongs to this step alone, and the clearance figure least of all. The cumulative XMuLV claim is a sum over 3 credited steps (Table 8.2), of which this one contributes 7.10 $\log_{10}$ and the anion exchange and virus filtration steps together contribute 11.77 $\log_{10}$. The drug substance aggregate figure reflects the cation exchange polish reported in PCR-007 as well as the increment added here, and the acidic charge variant figure is dominated by the level formed in the production bioreactor and reported in PCR-003.

Table 8.2: Modular viral clearance by step, with the cumulative claim.

step	XMuLV	MVM
Low-pH Viral Inactivation	7.1	0
Anion Exchange (AEX)	5.95	4.71
Virus Filtration	5.82	5.32
Cumulative	18.9	10

The robustness of this step's own contribution is best read from the Monte-Carlo analysis behind Table 7.1 rather than from the capability table. In that analysis each

parameter is scanned across its characterization range while the other two vary within their control bands. The predicted log reduction factor then stays above the required step contribution across the whole hold time range and the whole temperature range, and across the pH range up to 3.96. Two bounds apply, since the estimate assumes parameters inside their normal operating ranges and a qualified scale-down model, and it will be confirmed at commercial scale during Stage 2.

9 Parameter classification

Each parameter was classified against the decision logic of SOP-4001, which combines the demonstrated effect on a CQA with the risk that the parameter leaves the design space in routine operation. The outcome is given in the last column of Table 4.1 and the reasoning is set out below.

- Inactivation pH is a critical process parameter. It carries the largest effect on the attribute this step sets, its control band is narrow relative to the characterized range, and it is the parameter that defines the excluded corner of the design space. Values above the studied range have not been shown to inactivate XMuLV within the shortest acceptable hold, so the consequence of an excursion in the unfavourable direction is a loss of clearance rather than a loss of product quality.
- Hold time is a well-controlled critical process parameter. It has a significant effect on all three responses, and it is the only parameter that moves clearance and product quality in opposite directions. It is classified below pH because it is set by a timer against a controlled clock (the distributed control system record), its whole characterized range is proven acceptable for every attribute, and the risk of leaving the design space through hold time alone is very low.
- Temperature is a well-controlled critical process parameter. Its effect on clearance is significant but is the smallest of the three ($+0.64 \log_{10}$ across the whole characterized range), and its effect on aggregate is likewise smaller than that of hold time. It is held by a jacketed vessel within a band far narrower than the range studied.
- A-Mab concentration is a general process parameter. The univariate assessment covered the full range over which Protein A eluate is delivered, and no effect on a quality attribute governed by this step was demonstrated, so the whole delivered range is proven acceptable (Table 4.1). That null result is retained in the knowledge space as evidence that the step tolerates the whole delivered concentration range, and it is the reason the parameter needs no control beyond the routine monitoring already applied to the eluate.

Two further classification points follow from the results. No parameter of this step is a key process parameter, because none of them drives step yield or another process performance measure independently of the quality attributes above. The absence of any significant interaction in either design means that no parameter is classified on the strength of a combination effect. Every classification above rests on a main effect demonstrated in its own right.

10 Contribution to the control strategy

The step contributes four elements to the control strategy for the drug substance.

Parameter control is the first. Inactivation pH is controlled to the normal operating range in Table 4.1 and verified by an independent probe before the hold clock starts, which is the in-process control that protects the clearance claim. Hold time is controlled to its normal operating range with a minimum of 60 minutes, and temperature to its normal operating range. A-Mab concentration is monitored and not controlled to a range narrower than the eluate specification.

Record integrity is the second, and it arises from one of the deviations described below. The hold time recorded by the distributed control system is the controlled record for the step, the manual log is a secondary record, and clock synchronisation between the two is verified before execution. A discrepancy between them is treated as a deviation regardless of magnitude.

Monitoring and release testing are the third. The neutralised pool is tested for aggregate by SEC-HPLC (AMV-3011) and the drug substance is tested for charge variants by icIEF (AMV-3013), so both attributes this step affects are covered by release testing further down the process. Life-cycle control is the fourth, and it has two parts. The scale-down spiking model is requalified on the schedule set by SOP-1001, and the parameter ranges above enter continued process verification at commercial scale.

The step does not deliver viral safety on its own. Its XMULV increment is one of three credited contributions (Table 8.2), and it makes no contribution at all to the parvovirus claim, which rests on the anion exchange step (PCR-008) and the small-virus retentive filtration step (PCR-009). The enveloped-virus contributions are credited separately because their mechanisms are independent, and the cumulative claim is consolidated in PCMR-001. Aggregate control is shared with the cation exchange step (PCR-007), which reduces the level this step delivers.

11 Discussion

The step is well understood and its behaviour matches the mechanism assumed for it. The kinetic factors (pH, hold time and temperature) act additively on the log reduction factor, two of them act on aggregate, one of them acts on deamidation, and no interaction was found in either design. The operating region follows directly from those relationships, and the only boundary that binds is the clearance requirement at the corner where the highest pH meets the shortest hold and the lowest temperature. Prior knowledge from three earlier products predicted this structure, and the study confirmed it rather than revising it.

The adverse finding of the study is that corner, where 2 executed runs returned log reduction factors below the contribution required of the step. The fitted model agrees that the corner fails. The mitigating evidence is that the corner requires all three factors to be at their least favourable edge at once, that the normal operating ranges are far from it, and that the predicted margin at the least favourable corner of those ranges is 1.50 log₁₀. The residual position is that the corner is excluded from the design space and the step is not operated there.

Confidence in the scale-down model is high for the operating ranges and is necessarily qualified for the clearance claim. The hold is a homogeneous reaction in which the controlling variables are scale-independent, so the model reproduces the commercial step by construction and its qualification confirmed that it does. The clearance data are a different matter. They are generated with a laboratory virus preparation (a cultured XMuLV stock) spiked into the feed. The relationship between that preparation and an adventitious contaminant is an assumption of the method and not a result of this study (International Council for Harmonisation 2023a). That assumption is common to every modular clearance claim and is managed by the conservatism built into the requirement.

Three further limitations should be recorded. The response-surface model for the log reduction factor has a low predicted coefficient of determination, because the infectivity assay contributes a large part of the observed variance, and it should be read as a model of mean behaviour and not as a batch-level assurance statement. The design space has not been confirmed at commercial scale at the edges of the ranges, which is a general property of characterization work and is addressed by Stage 2. The acidic charge variant response returned no replicate variation in either design, so the study bounds the magnitude of the deamidation increment but provides no useful estimate of its variability.

The study recorded 2 deviations, and both were dispositioned as retained. Neither changed a fitted effect, a parameter classification or the operating region, and both are described in full in the deviations section below. One of them produced a lasting control, which is the record integrity element of the control strategy above.

Only one parameter of this step needs the most stringent control. Inactivation pH is classified as a critical process parameter, and hold time and temperature are well-controlled critical process parameters, which reflects the ease with which a timer and a jacket hold their targets compared with the consequence of a titration error.

12 Conclusions

The low-pH viral inactivation step is characterized and its operating region is defined. The step has 4 parameters, of which 3 were studied multivariately and 1 univariately. Inactivation pH is a critical process parameter, hold time and temperature are well-controlled critical process parameters, and A-Mab concentration is a general process parameter.

The response-surface models define a region covering the characterized ranges of pH, hold time and temperature, less the corner that combines the highest pH with the shortest hold and the lowest temperature. Within that region the XMuLV log reduction factor meets the step contribution of 4.93 log₁₀ required by the cumulative claim, the pool aggregate stays below 5 % and the acidic charge variants stay below 40 %. The normal operating ranges lie inside the region with a predicted clearance margin of 1.50 log₁₀ at their least favourable corner.

At commercial scale the cumulative XMuLV clearance has a one-sided Cpk of 1.53 and a margin of 2.81 log₁₀ to its requirement, which is the tightest of the attributes this step governs. The study recorded 2 deviations, and both were resolved without changing a fitted effect, a parameter classification or the reported operating region. The step assures a defined enveloped-virus inactivation increment within bounded product quality cost. It does not assure the viral safety of the drug substance by itself, and it makes no parvovirus claim. The findings roll up into PCMR-001.

13 Deviations from the plan

The study recorded 2 deviations during execution. Each was investigated, its root cause was established, its impact on the reported results was assessed, and both were dispositioned as retained. The register is given in Table 13.1 and each is described below.

Table 13.1: Deviations recorded during execution of PCR-006.

Devia- tion	Summary	Detected during	Disposi- tion
DEV-006-01	Hold-time record discrepancy between DCS and manual log	batch-record reconciliation	retained
DEV-006-02	XMULV infectivity assay cytotoxicity at low dilution	assay review	retained

Both entries in Table 13.1 were found after the affected runs were complete, one at batch-record reconciliation and one at assay review. Neither was detected by an in-process control at the time it occurred. The two post-execution checks that did find them are part of the study record, and both are described below.

13.1 DEV-006-01: hold-time record discrepancy

The recorded hold time for one run differed between two sources by 4 minutes. The distributed control system trend and the manual batch record disagreed, and the discrepancy was found during batch-record reconciliation after the run was complete. Until it was resolved, the hold time actually applied to that run was uncertain by that margin.

The investigation compared the timestamps of the two systems against a reference clock and found an offset between the distributed control system clock and the local panel clock used by the operator. The distributed control system record was confirmed as the accurate one and as the controlled record for the step. No equipment fault was found, and no other run of either design was affected.

The impact on the study is negligible on two grounds. The uncertainty of 4 minutes is small against the characterization range for hold time, which spans 60 to 180 minutes, so the run sits inside the studied range under either record. The effect of that uncertainty on the response is smaller still. Evaluating the fitted model at the set-point and again 4 minutes later changes the predicted log reduction factor by

0.024 $\log_{10}$, which is an order of magnitude below the residual standard error of the model (0.37 $\log_{10}$). The run was retained with the distributed control system value, and the fitted effects are unchanged. The deviation produced the record integrity control described in the control strategy section above.

13.2 DEV-006-02: infectivity assay cytotoxicity at low dilution

The sample matrix was cytotoxic to the indicator cell line at the lowest dilutions of the infectivity titration, so those wells could not be scored. The finding was made at assay review of the titration records. Titres for the affected samples were calculated from a dilution of one in 8 and above, following the cytotoxicity provisions of AMV-3017. Excluding the lowest dilutions raises the effective quantitation limit for those samples by 0.90 $\log_{10}$ above the validated limit of 1.5 $\log_{10}$ TCID₅₀/mL.

The root cause is the sample matrix itself. The post-hold samples are at low pH and carry the acid titrant, and both are toxic to the indicator cells at low dilution. This is a known property of low-pH hold samples and is the reason AMV-3017 carries a cytotoxicity control. No assay fault and no sample handling error were identified.

The consequence for the reported data is a conservative bias in one direction only. Where a post-hold titre could not be resolved below the raised quantitation limit, the log reduction factor reported for that run is a minimum, so the true clearance is at least the value used in the analysis. A bias of that sign cannot create an apparent clearance that does not exist, and it cannot mask a failing corner, since it can only make a reported value too small. The centre-point reproducibility of the assay across 4 replicate runs, at 0.38 $\log_{10}$, remains consistent with the validated intermediate precision of the method. The affected results were retained and the analysis was not repeated.

14 Appendices A-D

The appendices carry the full design matrices, the complete effect and coefficient tables for every response, and the analytical methods summary. Tables shown in the results sections above are not repeated here.

14.1 Appendix A — Screening design matrix and responses

The coded settings of the 11 screening runs are given in Table 14.1 and the measured responses in Table 14.2. Factor letters are defined in Table 4.2.

Table 14.1: Screening design, coded factor settings.

Run	Type	A	B	C
1	factorial	-1	-1	-1
2	factorial	-1	-1	1
3	factorial	-1	1	-1
4	factorial	-1	1	1
5	factorial	1	-1	-1
6	factorial	1	-1	1
7	factorial	1	1	-1
8	factorial	1	1	1
9	center	0	0	0
10	center	0	0	0
11	center	0	0	0

Table 14.2: Screening design, measured responses.

Run	Aggregates (HMW, %)	XMuLV LRF ($\log_{10}$)	Acidic variants
1	1.69	7.02	20.5
2	1.9	7.48	20.5
3	2.16	8.17	20.9
4	2.32	9.37	20.9
5	1.69	4.81	20.5
6	1.9	5.14	20.5
7	2.14	6.62	20.9
8	2.44	7.17	20.9
9	2.01	7.35	20.7

Run	Aggregates (HMW, %)	XMuLV LRF ($\log_{10}$)	Acidic variants
10	2.07	6.92	20.7
11	2.01	7.36	20.7

14.2 Appendix B — Response-surface design matrix and responses

The coded settings of the 18 response-surface runs are given in Table 14.3 and the measured responses in Table 14.4.

Table 14.3: Response-surface design, coded factor settings.

Run	Type	A	B	C
1	factorial	-1	-1	-1
2	factorial	-1	-1	1
3	factorial	-1	1	-1
4	factorial	-1	1	1
5	factorial	1	-1	-1
6	factorial	1	-1	1
7	factorial	1	1	-1
8	factorial	1	1	1
9	axial	-1	0	0
10	axial	1	0	0
11	axial	0	-1	0
12	axial	0	1	0
13	axial	0	0	-1
14	axial	0	0	1
15	center	0	0	0
16	center	0	0	0
17	center	0	0	0
18	center	0	0	0

Table 14.4: Response-surface design, measured responses.

Run	Aggregates (HMW, %)	XMuLV LRF ($\log_{10}$)	Acidic variants
1	1.69	6.49	20.5
2	1.88	7.35	20.5
3	2.17	7.65	20.9
4	2.36	7.91	20.9
5	1.68	4.82	20.5
6	1.88	5.77	20.5
7	2.11	6.45	20.9

Run	Aggregates (HMW, %)	XMuLV LRF (log ₁₀)	Acidic variants
8	2.39	7.16	20.9
9	2.01	7.04	20.7
10	2.02	6.13	20.7
11	1.79	6.65	20.5
12	2.24	7.96	20.9
13	1.92	7.22	20.7
14	2.13	7.29	20.7
15	2.02	6.94	20.7
16	1.98	6.15	20.7
17	2.02	6.97	20.7
18	2	6.78	20.7

14.3 Appendix C — Full effect and coefficient tables

The complete screening effect tables for all three responses are given in Table 14.5, Table 14.6 and Table 14.7. The response-surface coefficient table for the acidic charge variant response, which is not shown in the results sections, is given in Table 14.8. The diagnostics for the pool aggregate response-surface model are given in Figure 14.1.

Table 14.5: Screening effects on the pool aggregate level, all terms.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	0.471	0.236	0.0118	20	3.69e-05	***
C	0.221	0.11	0.0118	9.37	0.000723	***
AC	0.035	0.0175	0.0118	1.48	0.212	
AB	0.0285	0.0142	0.0118	1.21	0.294	
A	0.0221	0.011	0.0118	0.937	0.402	
BC	0.0107	0.00535	0.0118	0.453	0.674	

Table 14.6: Screening effects on the XMuLV log reduction factor, all terms.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	-2.08	-1.04	0.0936	-11.1	0.000376	***
B	1.72	0.862	0.0936	9.2	0.000774	***
C	0.636	0.318	0.0936	3.4	0.0274	*
BC	0.238	0.119	0.0936	1.27	0.272	
AB	0.203	0.101	0.0936	1.08	0.34	
AC	-0.194	-0.097	0.0936	-1.04	0.359	

Table 14.7: Screening effects on the acidic charge variant level, all terms. The estimates other than hold time are at the level of floating-point arithmetic and are not physical effects.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	0.4	0.2	4.12e-15	4.86e+13	1.08e-54	***
BC	-6.19e-15	-3.09e-15	4.12e-15	-0.751	0.494	
C	-5.35e-15	-2.67e-15	4.12e-15	-0.649	0.551	
AB	-4.57e-15	-2.29e-15	4.12e-15	-0.555	0.608	
AC	-2.6e-15	-1.3e-15	4.12e-15	-0.316	0.768	
A	6.69e-16	3.35e-16	4.12e-15	0.0812	0.939	

Table 14.8: Response-surface coefficients for the acidic charge variant level, all terms.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	20.7	3.39e-15	6.11e+15	5.79e-124	***
A	2.63e-15	2.72e-15	0.964	0.363	
B	0.2	2.72e-15	7.34e+13	1.32e-108	***
C	2.28e-15	2.72e-15	0.838	0.426	
AB	-1.19e-15	3.04e-15	-0.391	0.706	
AC	-2.19e-15	3.04e-15	-0.719	0.493	
BC	-3.12e-15	3.04e-15	-1.02	0.335	
A ²	-4e-15	5.23e-15	-0.764	0.467	
B ²	2.22e-15	5.23e-15	0.424	0.682	
C ²	-1.78e-15	5.23e-15	-0.34	0.743	

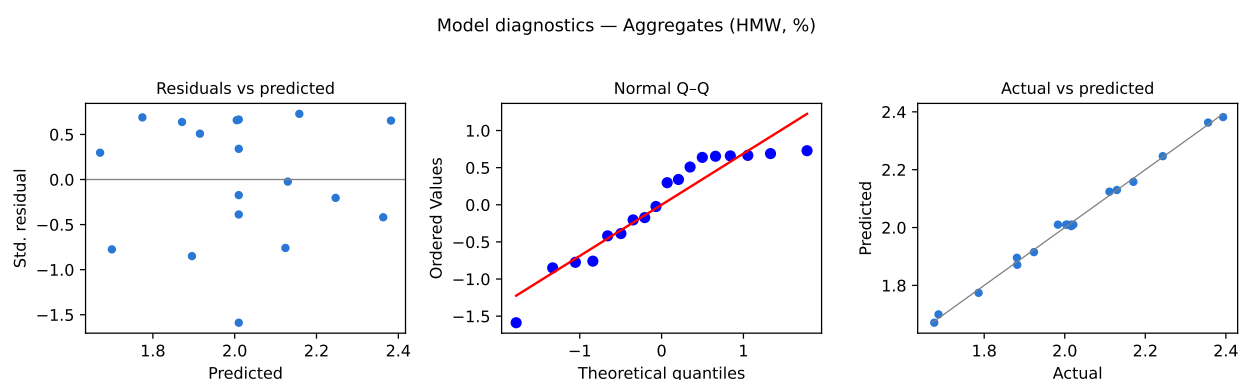


Figure 14.1: Model diagnostics for the pool aggregate response-surface model.

14.4 Appendix D — Analytical methods summary

The validated methods used in this study are summarized in Table 14.9. Method performance characteristics are held in the validation reports listed in Table 3.1.

Table 14.9: Analytical methods applied to the low-pH viral inactivation study.

Refer- ence	Technique	Analyte	Reportable value
AMV- 3017	Virus infectivity titration (TCID ₅₀)	XM _u LV	Log reduction factor (log ₁₀)
AMV- 3011	Size-variant analysis by SEC-HPLC	High molecular weight species	Per cent HMW
AMV- 3013	Charge-variant analysis by icIEF	Acidic charge variants	Per cent acidic

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